

Javier Lim | javier.lim@u.nus.edu | +65 8766 8771
javlim.dev | linkedin.com/in/jav-lim | github.com/javierlimt6

Availability: Aug. 2026 - Jul. 2027
San Francisco Bay Area

WORK AUTHORISATION

Singapore Citizen, eligible for **12-month J-1 US visa**, visa-sponsored by NUS, hosted by Stanford University.

EDUCATION

National University of Singapore | Bachelor of Computing in **Computer Science** 2027

NUS Scholar | Residential College Entrepreneurship Club **President** | Notion **Campus Leader** | **Teaching Assistant**

EXPERIENCE

ByteDance Singapore

Applied ML Orchestration Intern (ML Infrastructure) | k8s, Ray, Go, Scheduling, Training Apr. 2026 - Aug. 2026

- Building ML infrastructure for stage-aware autoscaling and resource optimisation across heterogeneous GPU/CPU clusters on Exabyte-scale training data for TikTok's global Search, Ads, & Recommendation models.
- Designing stage-aware autoscaling logic to detect distributed ML training lifecycle stages (warmup, failover, restart) and trigger lifecycle-aware, in-place scale-down strategies via a custom CRD-based controller.
- Implementing backpressure-driven VPA control across distributed training frameworks using throughput, GPU utilisation (SMA), and task lag to detect regressions & trigger automated rollback to prevent SLA degradation.
- Developing guaranteed in-place VPA scale-up for critical pods by integrating with the Kubernetes eviction layer to preemptively drain lower-priority workloads, leveraging QoS-class-aware eviction policies and admission webhooks.

Software Engineering Intern (BytePlus Platform Engineering) | Go, Node.js, RAG, Next.JS Jan. 2026 - Mar. 2026

- Engineered production features for byteplus.com supporting Seedream 5.0 launch, directly impacting international conversions for **1000+ B2B clients** & **440k+ MAU**.
- Leading 0→1 development of BytePlus **RAG** search: deploying **Go-driven Firecrawlers** for **ETL data ingestion of 5,000+ documentation pages**, integrated with SSE-supported **multimodal rendering** engine.
- Led performance engineering initiatives, implementing **in-memory caching** & experimental **edge computing** patterns to reduce SSR reliance & slash **TTI, LCP & FCP**, enhancing UX under high-server loads.

IBM (Client Engineering) Singapore

Full Stack Developer Intern | Spring Boot, Node.js, AWS, SQL Server, CrowdStrike, React Sep. 2025 - Dec. 2025

- **Forward deployed** on-site to co-engineer **B2G** SG military personnel **RBAC** system with 65k+ DAU.
- Built **Lambda**-automated event & visitor retry management batch pipeline using **Node.js** workers, integrating **FALCON** & Sequelize for clearance, triggering notifications using **AWS SNS/SES/SQS**.
- Implemented end-to-end conditional & automated on/offboarding with real-time **cross-system validation** to handle user **role management** across multiple locations, reducing manual data entry by **80%**.

CloudJoi (Malaysia's #1 arts ticketing platform) - cloudjoi.com Kuala Lumpur, Malaysia

Software Engineering Intern | Laravel, React Native, Next.js May. 2025 - Aug. 2025

- Contributed towards AI **semantic** search & recommendations via **vector embedding** & clustering metadata.
- Launched knowledgebase.cloudjoi.com—dashboard helpdesk for **200+ clients**—0→1 on **Next.js** with RESTful API.
- Deployed **ticket check-in/out system** for all live CloudJoi shows, enabling organisers to manage attendee status.
- Integrated **password change confirmation** flow with **queue-based email notifications** & Redis caching.

PROJECTS

Grid Universe AI Agent | PyTorch, NumPy, Scikit-learn Oct. 2025 - Nov. 2025

- Engineered an autonomous agent for a stochastic grid-world, across 48 complex test levels.
- Designed a custom CNN architecture (PyTorch) to reconstruct game state from snapshots with **99% accuracy**.
- Developed a lightweight statistical cipher decoder using SVD and NumPy to interpret encrypted objectives.
- Optimized A* search with Manhattan heuristics to handle dynamic hazards, achieving **100% success** in path finding.

TECHNICAL SKILLS

Languages: Go, Python, Java, TypeScript, PHP, JavaScript, Swift, C, C++, SQL, R, Assembly, Bash
Frameworks: React (Native), Node.js, Next.js, Laravel, Spring Boot, Supabase, Flask, Django, FastAPI
DevOps & Tools: Kubernetes, Ray, AWS, SQL Server, MySQL, MongoDB, PostgreSQL, Git, GCP, Docker
Libraries: Matplotlib, NumPy, pandas, SwiftUI, Three.js, sklearn, PyTorch